

Rishi Badal

945-276-4438 | dev.rishibadal@gmail.com | linkedin.com/in/rishi-badal | github.com/rishibadal726

Education

The University of Texas at Dallas

Bachelor of Science in Computer Science

Richardson, TX

Aug. 2025 – May 2029

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Systems Programming, Discrete Mathematics, Programming Fundamentals

Experience

Software Engineer

Nov. 2023 – Apr. 2024

NASA HUNCH Program

Allen, TX

- Earned **NASA HUNCH National Software Finalist** recognition by developing responsive front-end interfaces using **React, JavaScript, HTML, and CSS** for an environmental sensor subsystem monitoring temperature, humidity, and air-quality data in real time
- Designed monitoring, alarm, and controller interfaces using Figma and React to visualize real-time environmental readings and system alerts
- Improved the project through multiple design and competition stages by incorporating user testing, team feedback, and judge feedback within a 4-person team

Software Engineer

Sep. 2026 – Present

Nebula Labs

Richardson, TX

- Resolved account-update failures in a **Next.js, React, and TypeScript** application by synchronizing **BetterAuth** sessions through **trRPC**, preventing stale profile data after username changes
- Implemented profile-data handling that preserved user metadata during account updates, preventing invalid account information from being written to the **authentication system**
- Delivered code changes that passed linting, formatting, type checking, and build validation before being merged into the main codebase

Coding Tutor

Sep. 2026 – Present

Best Brains Learning Center

Anna, TX

- Instructed 5+ students in **Scratch, Python, and HTML/CSS**, developing their skills in block-based programming, scripting, and web development
- Guided students through coding exercises, debugging, and small projects across **3 classes per week**, strengthening problem-solving and independent programming skills

Projects

Lunar Lander | Python, PyTorch, Matplotlib, Gymnasium

Mar. 2026 – May 2026

- Trained a Deep Q-Network agent using **Python and PyTorch** across **2,000 training episodes** as part of a 7-person team, achieving a **75%+ landing success rate**
- Developed reward-tracking visualizations using Matplotlib that identified learning trends and crash penalties across repeated training runs
- Optimized training stability by storing up to **50,000 past experiences** for the agent to learn from and updating its learning model every **500 steps**, improving consistency across training runs

Shoppie | Python, TypeScript, React, Tailwind CSS, Supabase, Docker

Sep. 2026 – Present

- Developed an AI-powered shopping platform that centralizes product listings and prices across multiple retailers as part of an 8-person team
- Built the full-stack foundation using **Python, TypeScript, React, and Docker** to support product aggregation and comparison workflows
- Implemented product-comparison features that consolidate listings and prices from multiple retailers into a single shopping experience

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL (PostgreSQL), HTML/CSS

Frameworks: React, Next.js, Node.js, Spring Boot, FastAPI, Tailwind CSS

Developer Tools: Git, GitHub, Docker, Firebase, Supabase, Gemini API, Figma, VS Code

Libraries: PyTorch, NumPy, pandas, Matplotlib